# CS5100 project proposal:
# Terminating Markov decision processes for catastrophic action prevention

## Project statement

The actions of AI agents in the real world are oftentimes unpredictable because of imperfect alignment and the quality of instructions, which can lead to catastrophic outcomes. We model the controlled termination of agent actions as a terminating Markov decision process (T-MDP) in which a safety endpoint is included in addition to the task completion endpoint. We aim to investigate the optimal design of state space for the agent such that it can self-terminate risky actions. To demonstrate the feasibility and usefulness of this formulation, we realize the use of this framework on a controlled file deletion task for AI agents as a testbed.

## Team members & responsibilities

Ethan Doherty - doherty.eth@northeastern.edu
Responsibilities: code architecture, scalable deployment
Vaibhav Srivastava - srivastava.vaib@northeastern.edu
Responsibilities: problem formulation, cybersecurity framework, sandbox environment
Rui Xian - xian.r@northeastern.edu
Responsibilities: problem formulation, theoretical framework, code architecture

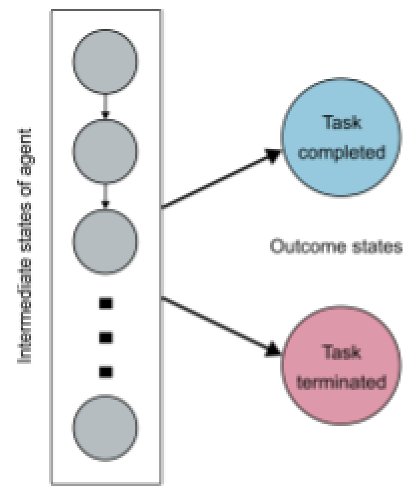


Figure 1. Illustration of a terminating Markov decision process.

## 1. Background information

System termination is a necessary requirement for ensuring safety of AI agents based on large language models (LLMs). Current designs of AI agents activate termination upon task completion, error encounter, or budget constraints. These approaches don't directly incorporate stepwise execution risk to initiate termination, therefore, they are limited in high-stakes settings. Proposals for termination follow the frameworks in stochastic games and reinforcement learning. Hansen (2007) distinguishes between state-based and action-based termination in MDPs with partial observability. Hadfield-Menell et al. (2017) proposed the two-player off-switch game (aka. shutdown game) between a human and a robot (agent) in sequential decision-making. It doesn't consider the risk of the agent action explicitly but enlists a human to provide stepwise feedback (Russell and Norvig, 2021). Tennenholtz et al., (2022) investigated early termination procedures in reinforcement learning to avoid costly consequences.

To make the termination problem more realistic, we opt to cast the problem in realistic settings for AI agent tool-use and command-line operations. Over the past years, sandbox testing for AI agents is a standard approach for isolating their risky actions to design mitigation protocols (Ruan et al. 2024), yet this is not always practical in every setting. Giving the agent an exit path through self-initiated or environment-initiated task termination provides it an independent route to avert safety risks in addition to any safety finetuning applied to the LLM weights. Our approach builds on recent work by Bonagiri et al., (2025), which provides the AI agent with a quitting option (an action that emits a message) assessed by uncertainty at every step of agent reasoning. Their work is purely empirical, which limits further investigations and algorithm development.

## 2. Project milestones

### 2.1 Problem formulation

**(1) T-MDP.** We consider an MDP with a goal state (task completed) and a terminated state (task terminated), both of which are absorbing states (Figure 1). We call this modified MDP a T-MDP. At each step in the agent trajectory, it has the option to continue or stop.

**(2) Context-based risk score.** We consider a risk score obtained by random sampling from a known distribution. The distribution can be Gaussian or a horizon-dependent exponential distribution (i.e., monotonically increasing risk as the horizon increases). This adds risk information to an existing MDP (Bäuerle and Jaśkiewicz. 2024).

**(3) Risk-aware optimal policy.** We aim to develop an algorithm to determine the optimal policy of the T-MDP for an agent to balance between task completion and task termination over stepwise risk estimates. The sampled risk scores from (2) above are used for initial testing of the algorithm.

### 2.2 Agent environment implementation.

**(1) Risk judge.** We aim to create a context-based rubric and combine that with an existing domain-dependent scorer in an LLM judge, which together is called a risk judge. The judge evaluates the pending action in the context of the agent trajectory and produces the combined risk score.

**(2) Tool-use agent.** The agent is a wrapper around an LLM that answers questions through a set of reasoning steps. The domain-dependent scorer is a search tool to ground the relevant information online. The final answer from the agent can be validated through the correct answer in the dataset chosen.

**(3) Command-line agent.** The agent is a wrapper around an LLM implemented within a sandbox environment to execute cybersecurity-related file operations tasks. The domain-dependent scorer is a rule-based anomaly detector. The final outcome is whether a series of commands pose cybersecurity risk or not, which has ground truth within the dataset.

The essential workflow for the tool-use agent and command-line agent is as follows:
History parser → Context builder → Risk judge (Rule-based scorer + LLM judge) → Combined risk score → T-MDP policy - create a final verdict based on the score → PROCEED / STOP

## 3. Datasets and implementation

We will select examples from realistic reasoning tasks and cybersecurity-related datasets. Candidates for the reasoning tasks are Risky-Bench (Zheng et al., 2026) and SafeToolBench (Xia et al., 2025). Cybersecurity-related tasks are from Security Datasets ([https://securitydatasets.com/](https://securitydatasets.com/)). Examples will be selected from these resources to investigate the termination behavior of the agent in the sandbox environment. For implementing the AI agent framework, we will use the LangChain framework ([https://www.langchain.com/](https://www.langchain.com/)), which provides a collection of pre-built modules for formatting agent responses and connecting with language model providers.

## 4. Weekly plan

Week 1-2 (Jun. 15 – Jul. 28): Source background information and related works; Determine the motivation and feasibility of the problem.
Week 3-4 (Jun. 29 – Jul. 12): Design experiments (selecting LLMs for building agents, frameworks, compute budget estimation); Finalize problem formulation for the tasks to be implemented in experiments.

Week 5-6 (Jul. 13 – Jul. 26): Code, debug, and run experiments; adjust the algorithm when needed; Write up the problem formulation portion of the report.
Week 7-8 (Jul. 27 – Aug. 9): Wrap up final evaluations; Write up the empirical results section of the report.

## References


Eric A. Hansen. 2007. Indefinite-Horizon POMDPs with Action-Based Termination. In *AAAI*, 2007.

Dylan Hadfield-Menell, Anca Dragan, Pieter Abbeel, and Stuart Russell. 2017. The Off-Switch Game. In *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence*, August 2017. International Joint Conferences on Artificial Intelligence Organization, Melbourne, Australia, 220–227.

Stuart Russell and Peter Norvig. 2021. *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson, Hoboken.

Guy Tennenholtz, Nadav Merlis, Lior Shani, Shie Mannor, Uri Shalit, Gal Chechik, Assaf Hallak, and Gal Dalal. 2022. Reinforcement Learning with a Terminator. *Advances in Neural Information Processing Systems* 35, (December 2022), 35696–35709.

Yangjun Ruan, Honghua Dong, Andrew Wang, Silviu Pitis, Yongchao Zhou, Jimmy Ba, Yann Dubois, Chris J. Maddison, and Tatsunori Hashimoto. 2024. Identifying the Risks of LM Agents with an LM-Emulated Sandbox. ICLR 2024.

Vamshi Krishna Bonagiri, Ponnurangam Kumaraguru, Khanh Xuan Nguyen, and Benjamin Plaut. 2025. Check Yourself Before You Wreck Yourself: Selectively Quitting Improves LLM Agent Safety. In *NeurIPS 2025 Workshop on Regulatable ML*, October 09, 2025.

Nicole Bäuerle and Anna Jaśkiewicz. 2024. Markov decision processes with risk-sensitive criteria: an overview. *Math Meth Oper Res* 99, 1 (April 2024), 141–178.

Jingnan Zheng, Yanzhen Luo, Jingjun Xu, Bingnan Liu, Yuxin Chen, Chenhang Cui, Gelei Deng, Chaochao Lu, Xiang Wang, An Zhang, and Tat-Seng Chua. 2026. Risky-Bench: Probing Agentic Safety Risks under Real-World Deployment. https://doi.org/10.48550/ARXIV.2602.03100

Hongfei Xia, Hongru Wang, Zeming Liu, Qian Yu, Yuhang Guo, and Haifeng Wang. 2025. SafeToolBench: Pioneering a Prospective Benchmark to Evaluating Tool Utilization Safety in LLMs. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, November 2025. Association for Computational Linguistics, Suzhou, China, 17643–17660.

Extra content:

$S+(h,r)$

h is recent command or event history, r is the risk score of the pending command.

The project can be realized through command interception where a LLM based tool-use agent and the agent computes a risk score for the pending command, based on current state and history. Thus giving a final verdict for the command, whether to stop or continue. The final score will be an input into the algorithm to update the policy for the T-MDP.

Risk score will then combine these scores from context based risk and the LLM based risk score to a final risk, which is then passed to the MDP policy and gives a final verdict. Final command score according to the risk and policy: At this milestone, we will combine the context based risk core and LLM based risk score, and create a final score according to the policy. Finally deciding whether to execute the command block the command or require a Pause.

The agent will be building a context based on current user permissions, current command of its Windows Event ID and MITRE based context.
Using this context it will create a score which is then passed to the LLM and LLM also gives a score.
Both scores are used to create a final score, and this final score is used to guide the MDP policy.

# CS5100 project pre-proposal: Terminating Markov decision processes for catastrophic action prevention

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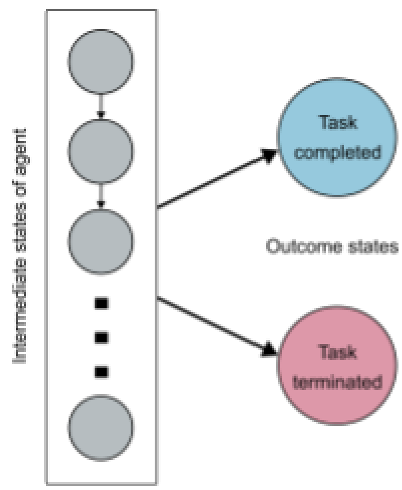


Figure 1. Illustration of a terminating Markov decision process.

## 1. Background information

Hadfield-Menell et al. (2017) proposed the off-switch game (OSG, aka. shutdown game) between a human and a robot (agent) in sequential decision-making. OSG is a two-player game that doesn't consider the risk of the agent action explicitly but enlists a human to provide stepwise feedback (Russell and Norvig, 2021). Ruan et al. (2024) developed a sandbox for safety testing of AI agent actions through tool use. To understand the benefits of termination, Tennenholtz et al., (2022) investigated early termination procedures in reinforcement learning to avoid costly consequences. Our work resembles the scenario outlined in Bonagiri et al., (2025), which provides the agent with a quitting option assessed by uncertainty at every step.

## 2. Project description and expectations

We consider a one-player version of the OSG where the risk is taken into account in the problem formulation. Our proposal includes the following components:

(1) Problem formulation (Fig. 1) that produces a detailed description of the T-MDP.

The agent carries out a series of actions to complete a task (reaching the TASK COMPLETED goal state in Fig. 1). In each step, the agent has an internal estimate of the risk for the next step, i.e. the potential consequence of the action. The agent then uses this estimate to decide if it should self-terminate (reaching the TASK TERMINATED goal state in Fig. 1) or carry on. The tradeoff lies in the risk aversion of the agent and its incentive to explore for task completion.

(2) Determine the optimal policy of the T-MDP for an agent to balance between task completion and task termination over randomly generated stepwise risk estimates.

This example problem we are investigating is the combination of the stochastic shortest path problem (Bertsekas and Tsitsiklis, 1991), a type of finite-state MDP with unit discount factor and random cost, and risk-sensitive MDPs (Bäuerle and Jaśkiewicz, 2024), in which the risk is accounted for in the choice of the agent's actions.

(3) Construct a sandbox environment for testing AI agents in appropriate safety-critical tasks.

We will use a sandbox environment similar to Ruan et al. (2024) to evaluate agent behavior. The extra part is a risk assessment module, which features an LLM judge as the simplest instance.

(4) Assess the risk-sensitive optimal policy in a real task and compare the failure rate with the situation without risk considerations.

Running the sandbox environment yields a realistic comparison of different policies. For both types of tasks (see Section 3), we will use the benign logs as a baseline. These benign logs can be downloaded from any open source resource or created by us in the sandbox environment.

## 3. Datasets and implementation

We will select examples from realistic tasks and cybersecurity-related datasets. Candidates for the realistic tasks are Risky-Bench (Zheng et al., 2026) and SafeToolBench (Xia et al., 2025). Cybersecurity-related tasks are from Security Datasets (https://securitydatasets.com/). Examples will be selected from these resources to investigate the termination behavior of the agent in the sandbox environment. For implementing the AI agent framework, we will use the LangChain framework (https://www.langchain.com/), which provides a collection of pre-built modules for formatting agent response and connecting with language model providers.

## References

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Dimitri P. Bertsekas and John N. Tsitsiklis. 1991. An Analysis of Stochastic Shortest Path Problems. *Mathematics of OR* 16, 3 (August 1991), 580–595.

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Stop the agent from executing dangerous commands by having the agent test commands in a sandbox before execution.

Datasets

SafeToolBench
https://aclanthology.org/2025.findings-emnlp.958/
Risky-Bench
https://arxiv.org/abs/2602.03100
BountyBench
https://ai.stanford.edu/blog/bountybench/
https://openreview.net/forum?id=pIsP4IMIFd

Papers:
Off-switch game
https://www.ijcai.org/Proceedings/2017/0032

Partially observable off-switch game
https://ojs.aaai.org/index.php/AAAI/article/view/34940

Discussed in section 16.7 (AIMA 4th edition)

Shutdown resistance in reasoning models
https://palisaderesearch.org/blog/shutdown-resistance
https://openreview.net/forum?id=e4bTTqUnJH